

Computational Musicology and Psychological Inference: A Methodological Framework for Trait and State Mapping

Theoretical Foundations of Musical Psychology and Preference Profiling

The assertion that musical choices provide a window into the human psyche is supported by decades of rigorous music psychology research. To transition from raw listening logs to a reliable psychometric profiling tool, a system must ground its assumptions in established taxonomies that map musical preferences to personality traits, cognitive thinking styles, and active emotion regulation strategies.

Personality and the MUSIC Model

Early attempts to link personality with musical taste were heavily constrained by broad, illusive, genre-based self-reports. This methodology failed to capture the stylistic fluidity of modern music consumption and suffered from social desirability bias. To address this limitation, Peter Rentfrow, Lewis Goldberg, and Daniel Levitin developed the MUSIC model, a latent, genre-free five-factor structure of musical preferences based on listeners' affective reactions to musical attributes.

The MUSIC model categorizes preferences into five primary meta-styles: Mellow, Urban, Sophisticated, Intense, and Unpretentious. Cross-cultural studies encompassing over 356,649 participants across 53 countries have demonstrated that the relationships between these meta-styles and the Big Five personality traits are globally invariant.

The "plasticity" traits—Openness to Experience and Extraversion—are the most powerful predictors of musical preference, whereas "stability" traits—Conscientiousness, Agreeableness, and Neuroticism—exhibit much weaker, though highly specific, correlations. Openness to Experience is the single greatest predictor of preference for Sophisticated and Intense styles. This correlation is driven by the aesthetic appreciation facet of openness, which attracts individuals to structural novelty and cognitive complexity.

Conscientiousness exhibits a strong, globally replicated negative correlation with Intense, rebellious genres such as heavy metal and punk rock. Neuroticism presents highly bifurcated preferences: some high-scoring individuals utilize Intense music to release inner angst and frustration, while others seek out Mellow music for mood-repair and comfort.

To enrich these behavioral baselines, Adrian North and Adrian Hargreaves formulated the "social badge" theory, demonstrating that adolescents and young adults actively employ music as an interpersonal badge to communicate values, status, attitudes, and self-views to peer groups. This makes the choice of high-visibility artists a strong indicator of social identity and self-congruity, which David Greenberg validated by showing that listeners prefer artists who

possess perceived personalities highly similar to their own self-congruity index.

MUSIC Meta-Style	Dominant Acoustic Attributes	Associated Big Five Traits	Cognitive & Social Functions	Archetypal Genres & Examples
Mellow (M)	Romantic, slow, quiet, acoustic, gentle, low energy.	Agreeableness, Openness to Experience.	Solace, emotional reflection, anxiety reduction.	Soft rock, R&B, quiet soul, singer-songwriter.
Urban (U)	Rhythmic, percussive, syncopated, vocal-heavy.	Extraversion.	Sociability, motor entrainment, social bonding.	Rap, hip-hop, funk, reggaeton, contemporary R&B.
Sophisticated (S)	Instrumental, harmonically complex, dynamic, intelligent.	Openness to Experience.	Cognitive stimulation, intellectual analysis, aesthetic "chills".	Classical, opera, avant-garde jazz, traditional world.
Intense (I)	Loud, aggressive, abrasive, high-arousal, distorted.	Low Conscientiousness, high Neuroticism sub-segments.	Catharsis, emotional discharge, rebellion, identity badging.	Heavy metal, punk rock, hard rock, alternative.
Unpretentious (U)	Uncomplicated, acoustic, relaxing, direct lyrical themes.	Conscientiousness, Agreeableness.	Stress reduction, passive distraction, stability.	Country, traditional acoustic pop, folk balladry.

Cognitive Styles and the Empathizing-Systemizing Theory

Beyond the Big Five, Greenberg and his colleagues linked musical preferences directly to cognitive styles using Simon Baron-Cohen's Empathizing-Systemizing (E-S) theory.

Empathizing (Type E) represents the drive to identify, predict, and respond to the emotional states of others. Systemizing (Type S) represents the drive to analyze, construct, and decode the underlying rules, patterns, and physical structures that govern non-agentive systems. Type B (Balanced) represents individuals who score relatively equally across both dimensions.

These cognitive styles correspond to highly distinct musical preference profiles. Empathizers prefer Mellow music featuring low acoustic energy, soft textures, and negative valence (sadness). In contrast, Systemizers show a strong preference for Intense and structurally complex music, such as hard rock, punk, avant-garde classical, and bebop jazz, which feature high arousal, cerebral depth, and intricate harmonic and rhythmic systems.

Neurobiological evidence supports these preferences. Systemizers possess larger gray matter volumes in prefrontal and cingulate regions implicated in error detection and cognitive control. Empathizers possess larger hypothalamic regions; when they listen to sad or mellow music, the acoustic features emulate sad human speech, triggering an empathetic response that stimulates the release of prolactin, emitting a consoling and soothing physiological effect.

Active Mood Regulation and the MMR Taxonomy

While personality and cognitive styles dictate long-term preference baselines, a listener's immediate state is governed by active emotion regulation. Suvi Saarikallio formulated the Music

in Mood Regulation (MMR) taxonomy, establishing seven distinct strategies that individuals utilize to manage their psychological states through music. These strategies, measured quantitatively via the 21-item Brief Music in Mood Regulation (B-MMR) scale, capture active, goal-directed musical interactions :

- **Entertainment:** Utilizing upbeat, pleasant music to maintain a positive, stable background atmosphere and sustain positive affect.
- **Revival:** Seeking out high-energy, rhythmic music to perk oneself up, overcome physical or mental fatigue, and gather new energy.
- **Strong Sensation:** Deliberately seeking out structurally intense or aesthetically profound music to induce powerful emotional states, bodily thrills, or aesthetic "chills".
- **Diversion:** Deploying music as a cognitive shield to actively distract the mind, forget unwanted thoughts, and redirect attention away from ongoing negative emotions.
- **Discharge:** Releasing internal negative emotions (e.g., anger, grief) by listening to music that explicitly expresses those same emotions, using the music as a coherent funnel for cathartic disclosure.
- **Mental Work:** Using music as a supportive cognitive framework for mental contemplation, problem-solving, and the reappraisal of emotional preoccupations.
- **Solace:** Seeking comfort, validation, and a sense of shared understanding when feeling sad, isolated, or troubled, treating the music as a consoling human presence.

Mechanisms of Emotional Elicitation and Behavioral Boundaries

Designing a computational profiling tool requires a deep understanding of *how* music evokes emotion, as well as the strict limits of inferring immediate psychological states from passive, naturalistic streaming logs.

Juslin's BRECVEMA Framework

To understand how sound is imbued with meaning and emotional resonance, Patrik Juslin developed the BRECVEMA model. This framework outlines eight distinct, evolutionary psychological mechanisms through which a musical event—defined by the unique interaction of the music, the listener, and the context—elicits an emotional response :

1. **Brain Stem Reflex:** A hardwired, subcortical response to fundamental acoustic properties (e.g., sudden loudness, high dissonance, accelerating tempos) that are interpreted by the brain stem as urgent, potentially life-threatening events.
2. **Rhythmic Entrainment:** A physiological alignment where a powerful, external musical tempo influences internal bodily rhythms (e.g., heart rate, respiration, motor cortex activity), locking them into a common periodicity. This is highly active in dance, march, and techno music.
3. **Evaluative Conditioning:** An associative process where a piece of music becomes paired with a positive or negative stimulus through repeated exposure, causing the music itself to eventually evoke the associated emotion in the absence of the original stimulus.
4. **Emotional Contagion:** An internal mirroring process where the listener perceives the expressive acoustic cues of the music (e.g., the crying, voice-like features of a violin or cello) and mimics them internally, inducing a corresponding emotional state.
5. **Visual Imagery:** An active cognitive process where the listener conjures up inner mental

images (e.g., a serene landscape or a past memory) in response to the musical structure, eliciting an emotional state from those visualized scenes.

6. **Episodic Memory:** A highly personalized process where a specific track acts as an associative key, unlocking a personal memory of a specific event or period in the listener's life, thereby re-arousing the original emotional state.
7. **Musical Expectancy:** A syntax-based cognitive process where the music violates, delays, or confirms the listener's expectations regarding the continuation of the musical structure, eliciting tension, surprise, or resolution.
8. **Aesthetic Judgment:** A higher-order cognitive appraisal where the listener evaluates the artistic, structural, or expressive value of the performance, generating "appreciation emotions" such as awe, wonder, and admiration.

The Sad Music Paradox and Clinical Risks

The phenomenon of enjoying music that evokes negative emotions is one of the most prominent paradoxes in musicology. Emery Schubert's "direct effect hypothesis" demonstrates that the removal of sadness from loved music directly decreases its aesthetic enjoyment for 82% of listeners, proving that the sadness itself contributes directly to the pleasure.

This emotional processing is determined by three parallel cognitive mechanisms: the Motivational Action Tendency (MAT), the Subjective Feeling (SF), and the Appraisal process. In real-world contexts, negative emotions couple these processes, meaning that feeling sad (SF) activates an aversive withdrawal or avoidance behavior (MAT). However, in an aesthetic context, the Appraisal process recognizes that the musical stimulus presents no real-world threat, decoupling the processes: the subjective feeling of sadness is experienced, but the aversive action tendency is inhibited, leaving the sadness to be enjoyed within a safe, simulated play environment.

While this decoupling allows for healthy, adaptive processing in most listeners, it poses a severe diagnostic hazard in others. Individuals with elevated tendencies toward maladaptive rumination frequently select sad music but fail to achieve decoupling. Instead of cognitive reappraisal, these ruminators experience a significant, measurable increase in depression, negative affect, and mood disturbance after listening.

Conversely, individuals scoring high in trait absorption (measured via the Modified Tellegen Absorption Scale) or reflectiveness successfully use sad music to process negative emotions, resulting in overall mood improvement.

An automated profiling tool must distinguish these states; simply tracking the consumption of sad music is insufficient, as the same stream log represents healthy emotional coping in a reflector, but an active, self-harming depressive spiral in a ruminator.

Behavioral Cohort	Primary Regulation Strategy	State vs. Trait Absorption Profile	Post-Listening Affective Impact	Algorithmic Risk of Misinterpretation
Healthy Reflector	Mental Work, Solace.	High trait absorption (MODTAS); highly variable state absorption (PCI).	Decoupled MAT; cognitive reappraisal, stress reduction, eventual positive mood repair.	None; safely mapped as healthy emotional processing and adaptive self-therapy.
Maladaptive	Solace-seeking,	Low-to-moderate	Coupled MAT;	High risk;

Behavioral Cohort	Primary Regulation Strategy	State vs. Trait Absorption Profile	Post-Listening Affective Impact	Algorithmic Risk of Misinterpretation
Ruminator	but devolves into cyclic, locked Solace.	trait absorption; high, obsessive state focus on negative stimuli.	increased depressive symptoms, elevated anxiety, and systemic mood disturbance.	algorithmic models will misinterpret excessive sad music plays as adaptive solace-seeking.
Background / Habitual Listener	Entertainment, Diversion.	Low trait absorption; negligible state absorption (focused on primary task).	Minimal immediate emotional impact; serves purely as background masking or distraction.	High risk; treating low-attention background streams as high-engagement emotional state updates.

Methodological Limits of State Inference

Isolating a listener's pure psychological traits from raw streaming data is complicated by several confounding factors:

- **Confounding Demographics:** Machine learning models that incorporate user demographics (age and gender) and general streaming habits (number of followed artists, device type) can achieve high personality prediction accuracy (correlations of $r = 0.26$ to $r = 0.37$). However, when these demographic variables are removed and classifiers are forced to rely strictly on raw acoustic features, prediction performance drops significantly. This indicates that much of what algorithmic models read as "individual psychology" is actually culturally and demographically normative behavior.
- **Feature Noise and Overfitting:** Combining disparate, high-dimensional feature spaces—such as low-level raw audio features, TF-IDF topic distributions, high-dimensional lyrical word embeddings, and structural segmentation tags—frequently introduces substantial feature noise, leading to model degradation. Because different features map to independent neurological processes, combining them indiscriminately can cause overfitting, and models trained on single, highly isolated feature groups (e.g., predicting Openness to Experience using *only* acousticness features) can outperform combined models.
- **Latency of Physiological Stress Reduction:** Objective physiological monitoring reveals that music does not reduce stress instantaneously. A minimum threshold of 20 minutes of continuous listening is required for physiological stress reduction to occur, and the effect is delayed, appearing in a time-lagged, delayed-latency manner. Furthermore, situational factors—such as whether the listener is alone or with others, and whether the music was selected with a specific intention to relax—strongly moderate the stress-reduction response. Tracking isolated, short-duration track plays (e.g., a single 3-minute relaxing track) as an instant state-repair event is therefore scientifically unsupported.

Prior Art Analysis of Commercial and Academic

Profiling Architectures

To build an expert-level profiling tool, one must evaluate the technical architectures, achievements, and shortcomings of both commercial and academic systems.

Commercial Year-End Summaries: Spotify Wrapped and Tinybird

Commercial year-end summaries are designed primarily as social-media-driven marketing campaigns to drive user retention and engagement. Despite their gamified aesthetics, they operate on complex, massive-scale data processing pipelines.

Spotify Wrapped and the Archive Engine

For the 2025 Wrapped campaign, Spotify deployed its "Wrapped Archive," generating 1.4 billion highly personalized reports for 350 million eligible users. The technical backend highlights the challenge of high-concurrency writes and distributed data aggregation :

- **Data Pipelines & Storage:** Raw listening logs are computed and pre-aggregated at the user level using a distributed data pipeline, exporting candidate records directly to object storage. To handle the massive, global traffic spike on launch day without write-lock bottlenecks, Spotify avoided serialized data schemas. They utilized a distributed, column-oriented key-value database designed for high-throughput writes, employing a schema where each remarkable day (up to five per user, such as "Biggest Discovery Day" or a day dominated by a single artist) was assigned its own column qualifier within a dedicated column family, using the date format YYYYMMDD as the unique qualifier.
- **Scale Mitigation via SMB:** To process petabytes of data efficiently, Spotify utilizes a "Sort Merge Bucket" (SMB) system, which groups and pre-sorts data into buckets based on key columns, significantly reducing the computational cost of join and aggregation operations while maintaining strict accuracy.
- **Storytelling via Fine-Tuned LLMs:** Once these remarkable days are isolated, their raw metadata logs (which include detailed track listings, artist counts, and overall user Wrapped statistics) are passed to a fine-tuned, frontier Large Language Model to generate witty, play-by-play narrative summaries. To evaluate these outputs at scale, Spotify utilized a parallel "LLM-as-a-judge" pipeline, grading the generated text across four distinct dimensions: accuracy, safety, tone, and formatting.
- **Shortcomings:** While technically impressive, Wrapped relies on highly rigid, gamified heuristics. Songs are ranked strictly by play counts, where a play is recorded only if a track is streamed for more than 30 seconds. Albums are only counted if the user has listened to at least 70% of its tracks, with ranking scores calculated based on play uniformity to assess true engagement. It ignores all music consumed outside of its own application (e.g., local files, physical media, radio). Furthermore, because users are aware of the impending annual event, many actively modify their listening habits in the preceding months to manipulate their final public-facing results, inducing data corruption.

Tinybird Operational Analytics Architecture

Tinybird's year-end "Wrapped" architecture highlights a highly efficient, real-time Lambda architecture designed to join historical pre-aggregated data with real-time streaming data :

- **Lambda Pattern execution:** The operational analytics pipeline joins pre-aggregated monthly historical data from January to November with real-time, daily operational metrics from December, allowing users to see their profiles update continuously up to the final seconds of the year.
- **ClickHouse Aggregation Engine:** Raw operational logs from multiple cloud regions are ingested into landing tables built on ClickHouse's ReplacingMergeTree engine, which automatically deduplicates late-arriving or overlapping batch metrics. Materialized views then perform continuous rollups of this raw data on an hourly, daily, and weekly basis. Crucially, these materialized views utilize ClickHouse's intermediate aggregation state functions, such as `sumMergeState()`, allowing the system to perform high-speed, multi-level rollups without rescanning the underlying raw tables. S3 Parquet serves as the interoperability layer, using auto-import connections to centralize multi-region metrics under a safe, public-facing, randomized identifier (`wrapped_id`), preventing the exposure of internal database organization IDs.

Event Logging and Academic Prototypes: Last.fm and MoodPlay

In contrast to the marketing-focused commercial tools, Last.fm and academic prototypes prioritize raw, continuous historical tracking and interactive emotional discovery.

Last.fm and Receiptify

Last.fm utilizes decentralized client wrappers to "scrobble" every track played across external players. It operates on a flat database model, displaying listening histories through unweighted play counts and simple timelines. While highly transparent, it possesses no native semantic understanding of emotional states or transitions. Receiptify operates similarly, acting as a lightweight wrapper for the Spotify/Last.fm APIs, translating top tracks into a retro shopping receipt. It utilizes absolute, unweighted frequency counts and contains no temporal decay, skip adjustments, or semantic features.

MoodPlay and myMoodplay

Developed as academic research prototypes by the Centre for Digital Music at Queen Mary University of London, MoodPlay and myMoodplay explored semantic web technologies, social tagging, and interactive emotional visualization :

- **System Architecture:** myMoodplay was built as a web-based music discovery app utilizing the Web Audio API, JavaScript animation engines, and Linked Data formats. Rather than relying on automated, low-level acoustic feature extractors, the system utilized crowd-sourced, social tag statistics mapped to a two-dimensional valence-arousal mood space to plot a track's coordinates. The client connected to a dedicated API, querying mood coordinates from a triple store using a SPARQL endpoint, and resolving artist metadata from external linked data sources.
- **Interactive trajectories:** Users navigated a large collection of over 10,000 songs by selecting desired coordinates on a continuous 2D touch interface, drawing a custom "mood trajectory path" that the system translated into a seamless, transition-optimized playlist.
- **Key Findings:** User evaluations revealed that crowd-sourced social tags and semantic metadata are significantly closer to human perceived emotion in music than

machine-extracted acoustic features. Furthermore, user experience testing demonstrated that a "mixed-mode" interaction model—combining natural language queries with intuitive, coordinate-based mood buttons—generated significantly higher user satisfaction than interfaces reliant on a single input mode.

Computational Aggregation, Recency Weighting, and Spatial Clustering Heuristics

To establish a mathematically rigorous methodology for a psychological profiling engine, developers must implement robust heuristics for temporal decay, active skip handling, and spatial density clustering.

Artist-Level vs. Track-Level Signals

The system must deploy a hybrid aggregation model that balances the high stability of artist-level signals with the high granularity of track-level acoustic features :

- **Artist-Level Vectors:** Artists carry heavy sociological, subcultural, and personal identity signals. By calculating an artist-level profile that incorporates self-congruity indices, the system can capture the listener's stable, long-term social identity, values, and personality alignment. In stream-count calculations, the primary artist should receive dominant weight, while featured artists receive supporting credit (typically $W_{\text{primary}} = 1.0$ and $W_{\text{featured}} = 0.3$), mirroring Spotify's established credit logic.
- **Track-Level Vectors:** Individual tracks capture immediate, highly variable emotional states and situational contexts. A mellow acoustic ballad and a high-tempo electronic remix by the same artist must be treated as highly distinct emotional events. Relying solely on artist-level centroids introduces severe emotional smoothing, obliterating these state transitions.

Dynamic Recency and Time-Decay Weighting

To capture both stable trait baselines and transient emotional shifts, the system should process listening events in parallel, utilizing a dynamic exponential time-decay function modeled on the human forgetting curve :

Where:

- $W_{ij}(t)$ is the mathematically adjusted weight of an interaction i with a track j at the current time t .
- W_0 is the raw interaction weight, determined by the engagement level of the stream:
- t_i is the exact epoch timestamp when the streaming event occurred, and t is the current epoch timestamp.
- λ is the exponential decay constant, dynamically derived from the target half-life window T_{half} :

By operating parallel processing pipelines with distinct half-life configurations, the system can isolate trait and state :

- **State Window ($T_{\text{half}} = 3 \text{ days}$):** Captures transient, highly repetitive, situation-specific listening patterns, mapping active mood regulation strategies, situational focus, or acute life stressors.
- **Trait Window ($T_{\text{half}} = 90 \text{ days}$):** Smooths out short-term contextual

noise, background listening anomalies, and intense situational hyper-fixations to isolate stable, underlying personality traits and cognitive styles.

Modeling Active Track Skips

Track skipping is one of the most powerful implicit feedback mechanisms in streaming data, representing an active aesthetic or emotional rejection. Analysis of commercial streaming databases shows that skipping is incredibly common: 25% of all streams are skipped within the first five seconds, and only 50% of tracks are played to completion.

The system must implement a strict, multi-tiered temporal skip-penalty heuristic based on user drop-off distributions and the Music Streaming Sessions Dataset (MSSD) :

Importantly, the system must cross-reference skips with the structural segmentation of the track. In commercial streaming, skipping likelihood is highly correlated with structural boundaries.

Listeners are highly prone to skip a song immediately following a transition between musical sections (e.g., right after an intro fades into a verse, or a chorus ends and a bridge begins).

A skip occurring precisely at a structural boundary indicates a natural, low-friction drop-off in attention, whereas a skip occurring mid-verse or mid-chorus represents a highly active, conscious aesthetic rejection. The model must apply a 0.5\times mitigation multiplier to the negative penalty of skips that fall within a \pm1.5-second window of a structural transition.

Density and Clustering Math in Emotional Space

To map the user's emotional state, played tracks are vectorized in a continuous, three-dimensional feature space representing Arousal, Valence, and Depth (the AVD model). Unsupervised clustering is then applied to locate high-density coordinates representing the user's emotional preoccupations.

Clustering Algorithm	Mathematical Mechanics	Advantages in Music Profiling	Critical Systemic Failures
K-Means Clustering	Partitions n data points into k distinct, non-overlapping clusters by iteratively minimizing the sum of squared Euclidean distances to the nearest cluster centroid.	Simple, computationally efficient, scales linearly with massive streaming datasets; produces highly balanced partitions of major lifestyle/mood states.	Requires pre-specifying k (e.g., using the Elbow method); forces outliers (e.g., accidental plays, background sleep music) into centroids, distorting the psychological profile.
DBSCAN	Groups points based on a local density threshold defined by neighborhood radius \epsilon and minimum points MinPts; unvisited points outside dense neighborhoods are classified as noise.	Automatically determines the optimal number of clusters; discovers arbitrary, organic shapes; isolates anomalous/accidental streaming events as noise.	Highly sensitive to parameter selection; fails completely when a user's data contains highly varying densities (e.g., massive, dense pop clusters alongside sparse classical exploration).
Gaussian Mixture	A probabilistic,	Soft clustering; assigns	Extremely sensitive to

Clustering Algorithm	Mathematical Mechanics	Advantages in Music Profiling	Critical Systemic Failures
Models (GMM)	generative model that assumes all data points are drawn from a mixture of a finite number of Gaussian distributions with unknown parameters.	continuous probabilities of cluster membership, allowing a single track to belong to multiple emotional and thematic profiles.	initialization parameters; high computational overhead, making real-time client-side execution difficult on massive logs.

The optimal clustering architecture is a **hybrid cascade**: the system should first execute DBSCAN to isolate and filter out outlier noise (e.g., background white noise, functional sleep playlists, or accidental streams). Once this noise is removed, the cleaned, high-density core data points are processed using a three-component Gaussian Mixture Model (GMM). This soft-clustering approach maps the overlapping boundaries of a user's core psychological states, preventing the artificial, hard boundaries of K-Means from mischaracterizing multi-functional tracks.

Automated Music Emotion and Lyrical Theme Classification as of 2026

The technological infrastructure of music information retrieval (MIR) has undergone a dramatic shift, transitioning from proprietary, centralized APIs to direct, independent audio processing and open data collectives.

The 2024-2026 Spotify API Disruption

On November 27, 2024, Spotify deprecated several of its most critical developer endpoints :

- GET /audio-features (BPM, key, mode, valence, energy, danceability).
- GET /audio-analysis (Structural segments, acoustic properties, timbre).
- GET /recommendations (Dynamic, seed-based recommendation generation).
- GET /related-artists.

This decision was driven by a dual strategy : securing the platform's proprietary data against the high-throughput scraping patterns of generative AI companies, and moving toward exclusive, monetization-focused integrations with professional DJ platforms (such as Rekordbox, Serato, and DJay, which launched privileged integrations across 51 markets on September 24, 2025). This API closure resulted in a massive developer backlash, prompting a structural rupture. On November 20, 2025, Anna's Archive released a 200GB compressed SQLite database containing complete metadata, audio features, and classifications for 256 million tracks and 186 million unique ISRCs, effectively democratizing the exact data Spotify sought to monopolize.

Modern API and Library Alternatives

In response to the Spotify deprecation, two distinct alternative ecosystems emerged :

Commercial Drop-In replacements (SoundNet / SoundStat)

SoundNet's Track Analysis API has emerged as the premier independent solution, performing

direct, server-side audio analysis on a rapidly growing database of over 6.7 million tracks. Accessible via a simple RapidAPI key and bypassing OAuth user authentication scopes, it returns a compact musical fingerprint that acts as a direct replacement for Spotify's deprecated features.

Open-Source and Crowd-Sourced Ecosystems (MetaBrainz)

ListenBrainz and MusicBrainz maintain open-source, community-driven datasets. ListenBrainz provides massive, AI-ready PostgreSQL data dumps of approximately 1 billion user "listens". MusicBrainz Picard uses the AcoustID audio fingerprinter—powered by the highly optimized Chromaprint algorithm—to generate compact fingerprints directly from raw audio files, allowing them to be matched to MusicBrainz IDs (MBID) and tagged with keys, BPM, and high-level classifiers, bypassing centralized streaming platforms entirely.

On-Premises Feature Extraction (Essentia)

For absolute independence from third-party APIs, developers utilize Essentia, an open-source C++/Python library developed by the Music Technology Group at Universitat Pompeu Fabra. Essentia offers a pre-compiled command-line extractor and advanced deep learning models :

- **Discogs-EffNet:** Audio embedding and classification models trained on Discogs metadata to predict 400 granular musical styles.
- **MusiCNN / VGGish:** Convolutional neural networks trained on the Million Song Dataset to predict high-level binary and multi-class categories (e.g., mood_happy with 81% accuracy, mood_sad with 86% accuracy, mood_acoustic with 90% accuracy, and voice_instrumental with 98% accuracy).
- **MAEST:** A state-of-the-art sequence transformer model trained with varying sequence lengths (5s, 10s, 20s, and 30s) initialized from pre-trained DeiT and PaSST weights, allowing for highly accurate temporal tracking of emotional shifts within a single song.

Lyrical and Theme Classification

Because lyrics represent a high-dimensional semantic signal that encodes a listener's narrative voice and thematic preferences independently of the music, accurate lyrical processing is essential.

Traditional NLP Failures

Traditional Vector Space Models (VSM) and standard TF-IDF Bag-of-Words pipelines are highly ineffective for lyrical analysis. Because song lyrics are structured differently than prose, they are heavily saturated with repetitive, non-contributing words (e.g., choruses, filler phrases) that introduce severe noise into standard classifiers. Furthermore, traditional VSM models are incapable of handling sentiment-reversing negations and intensity-altering modifiers. To overcome this, developers must deploy a Sentiment Vector Space Model (s-VSM), which filters out lexical noise by only extracting sentiment-bearing units and resolving nearby negations and intensifiers to generate a disambiguated emotional vector.

Deep Semantic and Transformer Frameworks

Modern lyrically-driven stylometric pipelines combine multiple layers of deep semantic analysis :

- **MPNet Embeddings:** Generate a 768-dimensional semantic vector that captures metaphorical meaning, narrative style, and deep contextual subtexts.
- **Zero-Shot Classifiers (BART-large-MNLI):** Generate probabilistic emotion scores across custom, non-pre-trained emotional classes (e.g., anger, joy, anxiety, calm) directly from raw text.
- **LyEmoBERT:** A specialized transformer model pre-trained on in-domain musical text, mapping lyrics to the four emotional quadrants of Russell's circumplex model of affect.

LLM-Based Thematic Coding and Systemic Biases

While LLMs (such as DeepSeek-V3, GPT-4o, or GLM4) are increasingly used for zero-shot playlist and lyric theme coding, systematic evaluations using the 5k-playlist PlaylistSense dataset have exposed severe limitations :

- **Comprehension Depth Bias:** LLMs exhibit strong performance when classifying surface-level, objective musical characteristics like "Music Culture" and "Music Era" (e.g., GPT-4o and GLM4 achieving high retrieval rates for 1980s synth-wave or 1990s hip-hop culture). However, their performance drops significantly when tasked with analyzing deep user psychology, complex emotional preoccupations, or abstract existential themes. Under evaluation, DeepSeek-V3 achieved a high 15.70% HR@1 score in Music Culture, but dropped to a negligible 0.88% HR@1 in Psychology. GLM4-9B-Chat achieved 17.71% in Music Culture, but only 1.50% in Psychology.
- **Cross-Dimensional Instability:** LLMs demonstrate significant performance fluctuations even within similar categories, making their zero-shot thematic classifications unstable and prone to hallucinations.

To build a reliable system, the thematic coding pipeline must bypass general-purpose LLMs for deep psychological profiling, utilizing instead specialized Large Musical Language Models (LMLMs) that are pre-trained directly on extensive lyrics-focused and music-criticism corpora.

Technical Tool	Capabilities & Output Features	Data Source & Access Method	Critical Limits & Vulnerabilities
SoundNet / SoundStat API	Key, major/minor mode, Camelot scale notation, BPM, energy, danceability, happiness, acousticness, instrumentality, speechiness, liveness, loudness.	Commercial API; RapidAPI simple key authentication; direct audio database analysis.	Paid pay-per-use structure; limited to standard Western-centric track structures; closed-source processing.
Essentia (MTG-UPF)	EBU R128 loudness, danceability, key/mode confidence, 400 Discogs subgenres, binary mood predictions (happy/sad/aggressive)	Local, on-premises execution; open-source C++ library with Python bindings.	Requires local computational resources; high memory overhead for running deep neural networks at scale.

Technical Tool	Capabilities & Output Features	Data Source & Access Method	Critical Limits & Vulnerabilities
ListenBrainz / MusicBrainz	Detailed user streaming timelines ("listens"), AcoustID Chromaprint matching, MusicBrainz ID resolution.	Fully open-source; PostgreSQL database dumps (~1 billion listens); open APIs.	Sparse feature coverage for highly niche, underground, or unreleased self-published music.
BART-large-MNLI / LyEmoBERT	Zero-shot emotional probabilities; 4-quadrant circumplex affective coordinates based on lyrical text.	Open-source Hugging Face models; on-premises PyTorch/Transformers pipeline.	Incapable of understanding instrumental tracks; highly sensitive to formatting, slang, and non-prose grammar.
General LLMs (DeepSeek-V3 / GPT-4o)	Zero-shot narrative writing, basic playlist clustering, surface-level thematic tag generation.	Cloud API access (OpenAI, DeepSeek); paid subscription or usage token model.	Pronounced Comprehension Depth Bias; struggles to understand deep user psychology and abstract emotions.

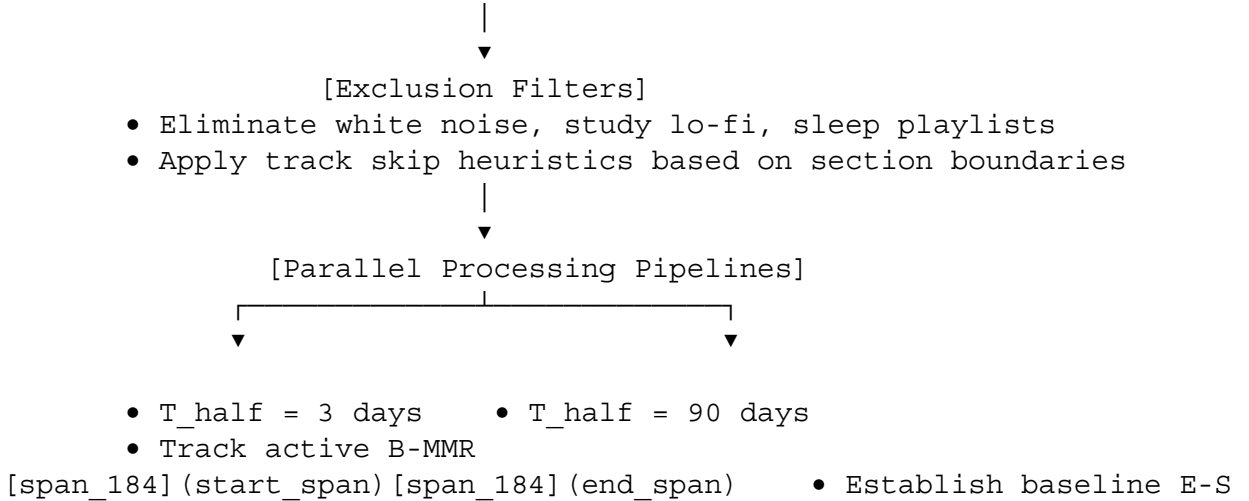
Analytical Synthesis and Actionable System Recommendations

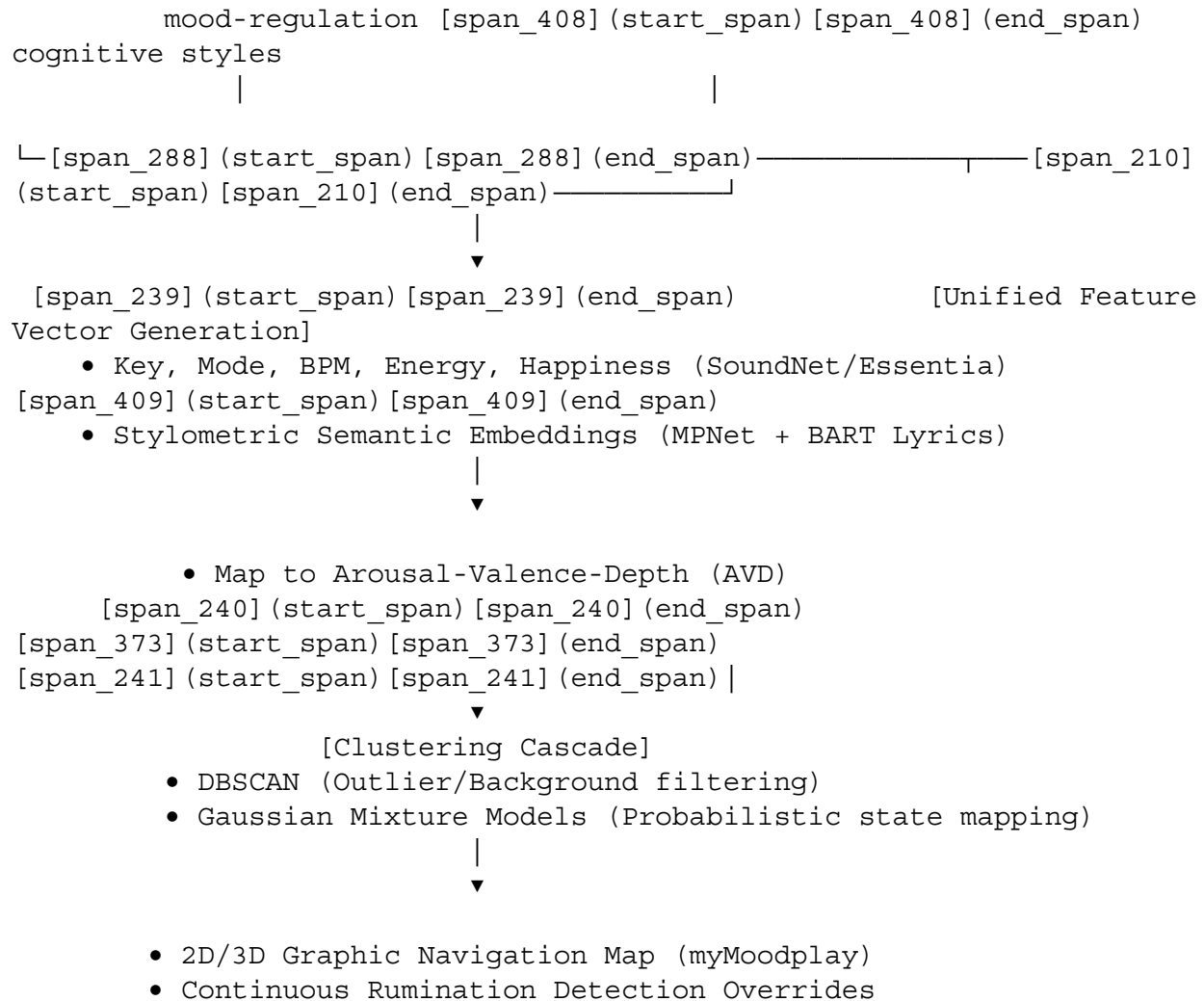
To build a production-grade system that surfaces patterns in a listener's emotional and thematic preoccupations, the following unified architecture is proposed:

1. Unified Dynamic Data Ingestion

The ingestion pipeline must support hybrid data sourcing to prevent on-platform bias. It should accept streaming history files from Spotify (utilizing extended historical data dumps) alongside continuous real-time scrobbles from ListenBrainz to capture cross-platform and physical media consumption.

Raw Listening Log ->





2. Temporal Aggregation and Recency Execution

The system should route incoming listening logs into the parallel Short-Term (State) and Long-Term (Trait) windows. For every session, compute interaction weights using the exponential decay formula:

Configure the short-term decay constant λ_{state} with a 3-day half-life to track volatile shifts, and the long-term decay constant λ_{trait} with a 90-day half-life to capture stable personality baselines.

3. Integrated Feature Extraction

For each unique track, pull features using a layered extraction model :

- **Acoustic Layer:** Query the SoundNet API to retrieve key, mode, BPM, energy, danceability, and happiness. If a track is missing from the database, trigger an on-premises Essentia extractor to generate MusiCNN mood predictions and EBU R128 loudness metrics.
- **Lyrical Layer:** For vocal tracks, pass the retrieved lyrics through a s-VSM processor to

isolate sentiment units. Run a zero-shot BART-large-MNLI classifier alongside MPNet semantic embeddings to generate high-dimensional theme vectors (e.g., isolation, nostalgia, achievement). Do not route raw lyrical summaries to general-purpose LLMs, as they suffer from Comprehension Depth Bias and will hallucinate incorrect psychological profiles.

4. Density Clustering Cascade

To map the user's emotional topography without distortion :

- **Step A:** Project all active track vectors into the 3D AVD space.
- **Step B:** Execute DBSCAN on the space. Any stream flagged as "noise" is discarded as background habituation, preventing functional background music from corrupting the emotional profile.
- **Step C:** Apply a three-component Gaussian Mixture Model (GMM) on the remaining cleaned coordinates. The soft, probabilistic boundaries of GMM will map the user's emotional landscape, representing their primary preoccupations as dynamic, overlapping density regions.

5. Clinical Safety and Rumination Overrides

To protect user wellbeing, the state analysis engine must implement an active clinical safety net :

- **Trigger condition:** If the 3-day state window detects a high-density cluster localized exclusively in the negative-valence, high-depth quadrant (Valence ≤ 0.3 , Depth ≥ 0.7) sustained continuously for more than 48 hours, activate the safety classifier.
- **Intervention:** Analyze the temporal trajectory. If the transition matrix shows zero progression toward higher-valence, comforting, or high-energy relief tracks, classify the state as a potentially harmful rumination loop rather than healthy solace-seeking.
- **Action:** The system must immediately adjust its recommendation engine, introducing a 0.5\times down-weight to the negative-valence state vectors and gently nudging the user's trajectory toward comforting, moderate-valence, low-arousal acoustic music (Solace-inducing Mellow tracks) to disrupt the rumination cycle.

6. Interactive Visual Graph Interface

Mirroring the successful interaction models of the academic myMoodplay prototype, the user interface must reject traditional, grid-based layout conventions. The tool should present the listener's emotional preoccupations as an organic, interactive 2D graph, where high-density emotional states are visualized as glowing, topographical cloud regions.

Users should be able to type natural language thematic queries (e.g., "processing nostalgic isolation") while simultaneously dragging a touch-cursor to draw customized trajectories across the visual emotional space, dynamically generating transition-optimized playlists that facilitate healthy, conscious emotional self-regulation.

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